Ratios, Proportions & Percentages

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Simplifying ratios**

Simplify the ratio $18 : 24$.

GCD of 18 and 24 is 6: $18 \div 6 : 24 \div 6 = 3 : 4$.

2 **[Foundational]** • **Target: Simplifying ratios**

Simplify the ratio $45 : 60 : 75$.

GCD of $45, 60, 75$ is 15: $3 : 4 : 5$.

3 **[Foundational]** • **Target: Simplifying ratios**

Simplify the ratio $36 : 8$.

GCD of 36 and 8 is 4: $9 : 2$.

4 **[Foundational]** • **Target: Simplifying ratios**

Express "12 boys to 18 girls" as a ratio in simplest form.

$12 : 18$, and GCD is 6, so $2 : 3$.

5 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Share 150 in the ratio $2 : 3$.

$2 + 3 = 5$ parts, each part $= 150 \div 5 = 30$. Shares: 60 and 90.

6 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Share 220 among three people in the ratio $3 : 4 : 4$.

$3 + 4 + 4 = 11$ parts, each part $= 220 \div 11 = 20$. Shares: $60, 80, 80$.

7 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

A prize of 900 riyals is shared between Sara and Layla in the ratio $5 : 4$. Find each share.

$5 + 4 = 9$ parts, each part $= 900 \div 9 = 100$. Sara receives 500 riyals; Layla receives 400 riyals.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Sharing a quantity in a given ratio

Divide 84 kg of flour in the ratio $1:2:4$.

$1+2+4=7$ parts, each part $=84\div 7=12$ kg. Shares: 12 kg, 24 kg, 48 kg.

9 [Intermediate] • Target: Equivalent ratios and proportions

If $3:5=x:20$, find x.

$\frac{3}{5}=\frac{x}{20}$, so $x=\frac{3\times 20}{5}=12$.

10 [Intermediate] • Target: Equivalent ratios and proportions

If $a:7=12:21$, find a.

$\frac{12}{21}=\frac{4}{7}$, so $a=4$.

11 [Intermediate] • Target: Equivalent ratios and proportions

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many meters in reality?

$3\times 50{,}000=150{,}000$ cm $=1{,}500$ m.

12 [Intermediate] • Target: Equivalent ratios and proportions

Solve for y: $\frac{4}{9}=\frac{y}{63}$.

$y=\frac{4\times 63}{9}=28$.

13 [Intermediate] • Target: Percent of a number

Find 35% of 240.

$\frac{35}{100}\times 240=84$.

14 [Intermediate] • Target: Percent of a number

Find 12% of 150.

$\frac{12}{100}\times 150=18$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Percent of a number • Cross-Lesson: 5.01 Quantitative Comparisons

What is 8% of 650?

$\frac{8}{100}\times 650=52$.

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16 [Challenging] • Target: Percent of a number • Cross-Lesson: 6.01 Sequences & Patterns

Find 150% of 40.

$\frac{150}{100} \times 40 = 60$.

17 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 7.01 Data Analysis & Probability

Write $\frac{3}{8}$ as a percent.

$\frac{3}{8} = 0.375 = 37.5\%$.

18 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 8.01 Word Problems & Reasoning

Write 0.45 as a fraction in lowest terms.

$0.45 = \frac{45}{100} = \frac{9}{20}$.

19 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 1.01 Arithmetic & Number Sense

Write 62.5% as a fraction in lowest terms.

$62.5\% = \frac{625}{1000} = \frac{5}{8}$.

20 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 3.01 Algebra Essentials

Write $\frac{7}{40}$ as a decimal.

$\frac{7}{40} = 0.175$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Sharing in a given ratio

The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.

a. 36

b. 21

c. 48 ✓

d. 63

Total parts $= 3 + 4 = 7$; each part $= 84 \div 7 = 12$. Larger share $= 4 \times 12 = 48$.

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22 [Foundational] • Target: MCQ — Percent of a number

The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?

a. 68

b. 64

c. 60 ✓

d. 56

75% of 80 = 0.75 × 80 = 60.

23 [Foundational] • Target: MCQ — Percent of a number

The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?

a. Apt. 2

b. Apt. 1

c. **Apt. 3** ✓

d. Apt. 4

0.08 × 2,500 = 200 SAR, matching Apartment 3's rent of 2,500 SAR.

24 [Foundational] • Target: MCQ — Percentage increase & decrease

The bar chart shows a salary before and after a raise. What percentage increase does it represent?

a. 15%

b. 500%

c. 20% ✓

d. 25%

Increase = 6,000 − 5,000 = 1,000. Percentage increase = $\frac{1,000}{5,000}$ × 100% = 20%.

25 [Foundational] • Target: MCQ — Successive percentage changes

The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.

a. 500

b. 900

c. 1,000

d. 750 ✓

1,000 × 1.5 × 0.5 = 750.

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26 [Foundational] • Target: MCQ — Direct proportion

The graph shows the direct proportion between the number of hours x worked and pay y in SAR, $y = 15x$. How much is earned for 6 hours?

a. 105 **b.** 90 ✓

c. 75 **d.** 21

$y = 15 \times 6 = 90$ SAR.

27 [Foundational] • Target: MCQ — Best value & unit price

The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?

a. **Brand B** ✓ **b.** All are equal

c. Brand C **d.** Brand A

The lowest price per 100 g is the best value — Brand B at 9 SAR/100g beats Brand A (12) and Brand C (11).

28 [Foundational] • Target: MCQ — Best value & unit price

The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?

a. Cannot be determined **b.** They are equal

c. **The 600 g box** ✓ **d.** The 400 g box

400 g: $12 \div 4 = 3$ SAR/100g. 600 g: $16.50 \div 6 = 2.75$ SAR/100g. The 600 g box is cheaper per 100 g.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Simplifying & writing ratios

Simplify the ratio $42 : 56$.

a. $3 : 4$ ✓ **b.** $6 : 8$

c. $7 : 8$ **d.** $3 : 8$

Divide both terms by their GCD, 14: $42 \div 14 : 56 \div 14 = 3 : 4$.

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30 [Intermediate] • Target: MCQ — Sharing in a given ratio

Two business partners share a profit of $18{,}000$ SAR in the ratio $5 : 7$. Find the larger share.

a. $10{,}500$ **SAR** ✓

b. $9{,}000$ SAR

c. $7{,}500$ SAR

d. $12{,}857$ SAR

Each part $= 18{,}000 \div 12 = 1{,}500$. Larger share $= 7 \times 1{,}500 = 10{,}500$ SAR.

31 [Intermediate] • Target: MCQ — Equivalent ratios & scale

Which ratio is equivalent to $4 : 9$?

a. $9 : 4$

b. $20 : 45$ ✓

c. $8 : 16$

d. $16 : 37$

$4 : 9$ scaled by 5 gives $20 : 45$.

32 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Write $\frac{7}{20}$ as a percent.

a. 35% ✓

b. 3.5%

c. 7%

d. 70%

$\frac{7}{20} = \frac{35}{100} = 35\%$.

33 [Intermediate] • Target: MCQ — Percentage increase & decrease

A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 **SAR** ✓

b. 144 SAR

c. 100 SAR

d. 140 SAR

$120 = 0.8 \times \text{original} \Rightarrow \text{original} = 120 \div 0.8 = 150$ SAR.

34 [Intermediate] • Target: MCQ — Finding the original value

After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.

a. 158.70 SAR

b. 120 **SAR** ✓

c. 123 SAR

d. 117.30 SAR

$138 = 1.15 \times \text{original} \Rightarrow \text{original} = 138 \div 1.15 = 120$ SAR.

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35 [Intermediate] • Target: MCQ — Direct proportion

A car uses 8 liters of fuel to travel 96 km. At the same rate, how far can it travel on 20 liters?

a. 200

b. 260

c. 240 ✓

d. 180

Rate $= 96 \div 8 = 12$ km per liter. $20 \times 12 = 240$ km.

36 [Intermediate] • Target: MCQ — Best value & unit price

A 2 L bottle of juice costs 18 SAR. Find the price per 100 mL.

a. 9 SAR

b. 0.18 SAR

c. 1.80 SAR

d. 0.90 **SAR** ✓

2 L $= 2{,}000$ mL. $18 \div 2{,}000 \times 100 = 0.90$ SAR per 100 mL.

37 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Find 15% of 160.

a. 19

b. 29

c. 36

d. 24 ✓

$\frac{15}{100} \times 160 = 24$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Percentage increase & decrease • Cross-Lesson: 8.01 Word Problems & Reasoning

Find 10% of 400.

a. 52

b. 40 ✓

c. 35

d. 45

$\frac{10}{100} \times 400 = 40$.

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39 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find 5% of 100.

a. 5 ✓

b. 17

c. 10

d. 25

$\frac{5}{100} \times 100 = 5$.

40 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 3.01 Algebra Essentials

Decrease 210 by 10%.

a. 185

b. 189 ✓

c. 198

d. 193

$210 \times 0.9 = 189$.

41 [Challenging] • Target: MCQ — Direct proportion • Cross-Lesson: 4.01 Geometry & Measurement

Increase 310 by 10%.

a. 350

b. 345

c. 341 ✓

d. 337

$310 \times 1.1 = 341$.

42 [Challenging] • Target: MCQ — Inverse proportion • Cross-Lesson: 5.01 Quantitative Comparisons

Decrease 340 by 50%.

a. 166

b. 179

c. 174

d. 170 ✓

$340 \times 0.5 = 170$.

43 [Challenging] • Target: MCQ — Best value & unit price • Cross-Lesson: 6.01 Sequences & Patterns

Share 390 among three people in the ratio 7 : 3 : 3.

a. 210, 90, 90 ✓

b. 90, 90, 210

c. 240, 90, 90

d. 210, 90, 60

$7 + 3 + 3 = 13$ parts, each part $= 390 \div 13 = 30$. Shares: 210, 90, 90.

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44 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 7.01 Data Analysis & Probability

7 liters of paint cost 98 riyals. Find the cost of 5 liters of paint at the same rate.

a. 80

b. 56

c. 70 ✓

d. 84

Cost per unit $= 98 \div 7 = 14$ riyals. Cost of 5: $14 \times 5 = 70$ riyals.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Word Problems & Reasoning

After a 50% increase, a quantity is now 90. Find the original value.

a. 60 ✓

b. 70

c. 50

d. 90

Original $\times 1.5 = 90$, so original $= 90 \div 1.5 = 60$.

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